

Nathan O'Brien

Active Secret Clearance | U.S. Citizen | (904) 518-9264 | nathanobrien261@gmail.com | linkedin.com/in/nathano-brien

Education

Georgia Institute of Technology | Atlanta, GA

August 2025 – Present

M.S. Aerospace Engineering | GPA: 4.00 / 4.00 | NSSA RGi Scholarship

University of Florida | Gainesville, FL

Graduated December 2024

B.S. Aerospace Engineering | GPA: 3.86 / 4.00 | Presidential Platinum Scholarship

Professional Experience

Draper | Houston, TX

June 2026 – August 2026

Guidance, Navigation & Controls Engineering Intern

- Evaluated the Orion Crew Capsule's Atmospheric Extended Kalman Filters in the presence of IMU power faults and channel overrides to determine the absolute latest time that an IMU could be safely power cycled on the pad.
- Conducted Monte Carlo analysis in the TRICK simulation environment to verify that state transfer commands were correctly updating exponentially correlated random variables used to model sensor error states.
- Created a CICD pipeline component used to identify discrepancies in time handling between GNSS measurements and buffered accelerometer measurements, resulting in correction of several faulty time labelling functions.

York Space Systems | Denver, CO

January 2025 – July 2025

Attitude Control Systems & Mission Operations Intern – Secret Clearance

- Utilized alignment quaternions and MATLAB to create a 3D rendering tool used to visualize satellite component coordinate systems in reference to one another during various simulated scenarios and assembly processes.
- Trained shop technicians, mission operators and attitude control engineers to use the rendering tool, resulting in identification of torque rod and sun sensor misalignments as well as correction of sensor obfuscation issues.
- Acted as a certified Vehicle Ground Controller for experimental Proliferated Warfighter Space Architecture military communications satellites and commanded on-orbit testing of optical communications terminals.

Blue Origin | Kent, WA

September 2023 – December 2023

Guidance, Navigation & Controls Engineering Intern

- Generated 50+ MATLAB unit tests to automate checks of IMU, GNSS receiver and star tracker settings and performance metrics between simulated New Glenn trajectories during processor-in-the-loop testing campaigns.
- Traced navigational data flows to update Confluence documentation regarding how position, velocity and attitude were calculated throughout flight using interview testimony from senior engineers and internal whitepapers.

Extracurricular Experience

Engineering Space Policy Laboratory | PI: Professor Thomas González Roberts

November 2025 – Present

Lead Space Security Graduate Researcher

- Authored a Python GitHub repository used to generate continuous-coverage space-based interceptor constellations.
- Utilized a mixed integer linear programming method to optimally select orbits given target region(s) and salvo size.
- Created a batch runner framework to automatically trigger runs and store results using AWS cloud functionality.

Florida Rocket Lab Student Design Team

July 2021 – December 2024

Business Development Director

- Founded a marketing subteam that increased the team budget by \$45,000 using a sponsorship incentive structure.
- Directed development of the team website and social media accounts to network with commercial aerospace firms.
- Presented to former members of the US Congress regarding commercial space engagement on college campuses.

Technical Skills

MATLAB | Simulink | Python | Git | Claude Code | AWS | Gurobi | LabView | Solidworks | Jira | Confluence | MSOffice

Graduate Coursework

Kalman Filtering | Optical Navigation | Spacecraft Attitude Estimation | Space Policy | Optimal Guidance and Control
Linear Control Systems | Math Principles of Autonomy | Space Sustainability | Astrodynamics | Sensor-Based Planning